Chapter 5
Continuous-Time Markov Chain

- Definition
- Birth-death process
- The Q matrix
- Limiting probability

Definition

- Suppose we have a continuous-time stochastic process $\{X(t), t \geq 0\}$ taking on values in the set of nonnegative integers. We say that the process $\{X(t), t \geq 0\}$ is a continuous-time Markov chain (CTMC) if for all $s, t \geq 0$ and nonnegative integers i, j

$$P(X(t+s) = j | X(s) = i, X(u) = x(u), 0 \leq u < s) = P(X(t+s) = j | X(s) = i).$$

future now past

- If in addition, the conditional probability $P(X(t+s) = j | X(s) = i) = P(X(t) = j | X(0) = i)$ for all $s \geq 0$, we say that $X(t)$ has stationary or time-homogeneous transition probability.
- In this chapter, we shall study only time-homogeneous CTMC.

Definition

- **Fact:** If $X(t)$ is a CTMC and $X(0) = i$, let H_i be the amount of time $X(t)$ spends in state i before it leaves state i. (H_i is called the holding time.) Then, H_i must follow an exponential distribution.

Alternative Defintion

- A CTMC is a stochastic process having the properties that each time it enters state i,
  (a) The amount of time it spends in that state before making a transition into a different state is exponentially distributed with mean, say, $1/v_i$;
  (b) When the process leaves state i, it next enters state j with some probability, say, P_{ij}.

- In this term, CTMC can be derived from a DTMC by adding exponential inter-transition times.
- The corresponding DTMC is called the embedded chain.

Examples

- **Example 6.1 (A Shoe Shine Shop).** Consider a shoe shine establishment consisting of two chairs-chair 1 and chair 2. A customer upon arrival goes initially to chair 1 where his shoes are cleaned and polish is applied. After this is done the customer moves on to chair 2 where the polish is buffed. The service times at the two chairs are assumed to be independent random variables that are exponentially distributed with respective rates μ_1 and μ_2. Suppose that potential customers arrive in accordance with a Poisson process having rate λ, and that a potential customer will enter the system only if both chairs are empty.

To describe the MC, let the states be $\{0, 1, 2\}$ such that

State	Interpretation
0	System is empty
1	a customer in chair 1
2	" " " 2

- Starting at 0 and will move to 1 if a customer comes, the holding time is $Exp(\lambda)$.
- Starting at 1 and will move to 2 with time $Exp(\mu_1)$.
- Starting at 2 and will move to 0 with time $Exp(\mu_2)$.

In Summary.

State	*Interpretation*
0	system is empty
1	a customer is in chair 1
2	a customer is in chair 2

We leave it as an exercise for you to verify that

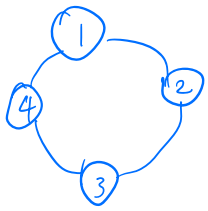
$$v_0 = \lambda, \quad v_1 = \mu_1, \quad v_2 = \mu_2,$$
$$P_{01} = P_{12} = P_{20} = 1$$

The embedding chain is $S = \{0, 1, 2\}$ with transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Examples

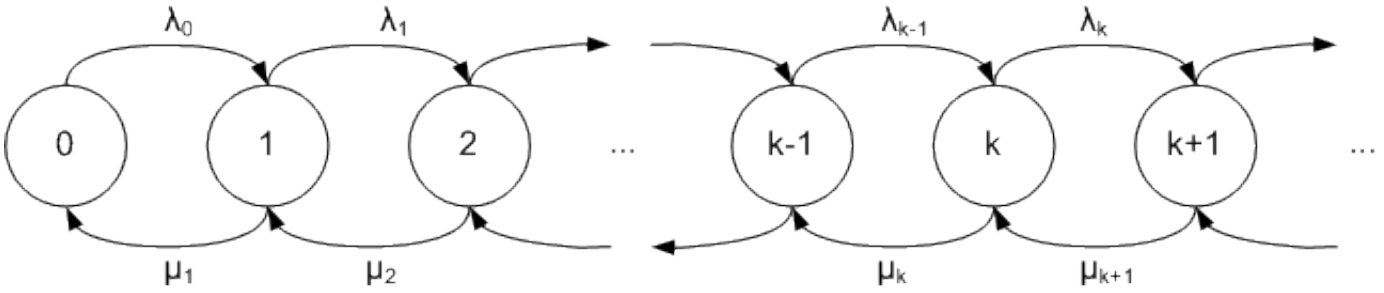
- **Example (A Rat in Maze)** Consider a rat in the closed maze, in which at each transition, the rat is equally likely to move to one of the neighboring two cells, but where now we assume that the holding time, H_i , in cell i is exponential at rate a_i, $i = 1, 2, 3, 4$. Time is in minutes. Let $X(t)$ denote the cell that the rat is in at time t. $\{X(t)\}$ forms a CTMC.



$P_{ij} = \frac{1}{2}$ for any i to any of its neighbors

Holding time at $i \sim \text{Exp}(a_i)$.

Birth-Death Process



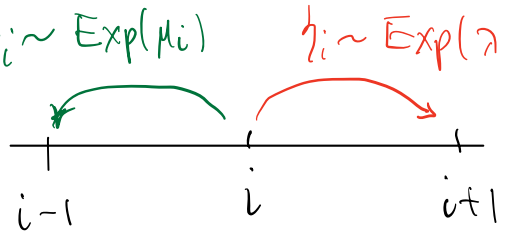
- A birth-death (BD) process is a CTMC such that the chain can only move from state i to state $i+1$ or $i-1$ in one transition.

H_i : the holding time at i with $Exp(\nu_i)$-dist.

- Question: what is the distribution of H_i?

- Example: Poisson process is a BD process with $\mu_n = 0$ and $\lambda_n = \lambda$.

Solution.



$H_i = \min\{\xi_i, \eta_i\}$, where ξ_i and η_i are indept.

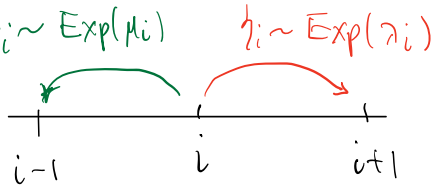
Thus, $H_i \sim Exp(\mu_i + \lambda_i)$ for $i \geq 1$.

$$P_{i\,i+1} = P(\eta_i < \xi_i) = \frac{\lambda_i}{\lambda_i + \mu_i}.$$

$$P_{i\,i-1} = P(\xi_i < \eta_i) = \frac{\mu_i}{\lambda_i + \mu_i}, \quad i \geq 1.$$

For $i = 0$, $P_{01} = 1$. As above, $P_{10} = \frac{\mu_1}{\lambda_1 + \mu_1}$.

In summary, a birth-death process results with a Markov process with transition matrix $P = (P_{ij})_{i,j \geq 0}$. Find the exponential dist's at each location.

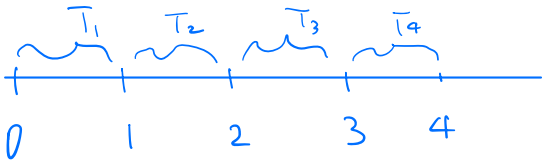


$$P_{i\,i-1} = \frac{\mu_i}{\mu_i + \lambda_i}, \quad P_{i\,i+1} = \frac{\lambda_i}{\mu_i + \lambda_i}, \quad V_i = \lambda_i + \mu_i$$

Since $P_{01} = 1$, then $\mu_1 = 0$, λ_1 is arbitrary.

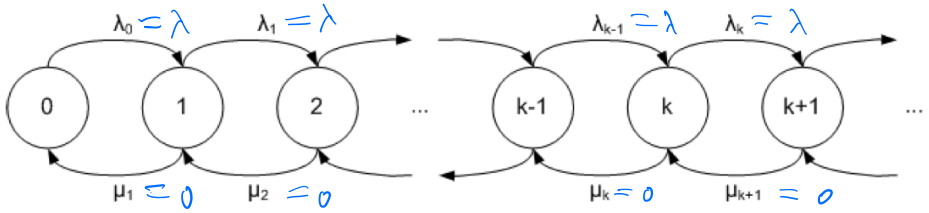
For $i \geq 1$, $\mu_i \equiv \frac{P_{i\,i-1}}{1 - P_{i\,i-1}} \lambda_i$, λ_i is arbitrary.

Example. Poisson process



Interarrival times: $T_1, T_2, \cdots$ are iid $Exp(\lambda)$.

$N(t)$ = # of occurence of events up to time t.



$\lambda_i \equiv \lambda$.

$\mu_i \equiv 0$.

Examples

- **Example 6.4 (A Linear Growth Model with Immigration)** A model in which

$$\mu_n = n\mu, n \geq 1, \lambda_n = n\lambda + \theta, n \geq 0$$

is called a linear growth process with immigration. Such processes occur naturally in the study of biological reproduction and population growth. Each individual in the population is assumed to give birth at an exponential rate λ; in addition, there is an exponential rate of increase θ of the population due to an external source such as immigration. Hence, the total birth rate where there are n persons in the system is $n\lambda + \theta$. Deaths are assumed to occur at an exponential rate μ for each member of the population, so $\mu_n = n\mu$.

Examples

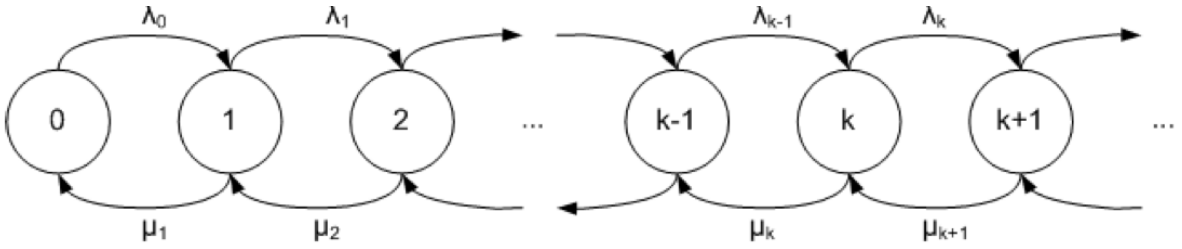
- **Example 6.5 (The Queueing System M/M/1)** Suppose that customers arrive at a single-server service station in accordance with a Poisson process having rate λ. That is, the times between successive arrivals are independent exponential random variables having mean $1/\lambda$. Upon arrival, each customer goes directly into service if the server is free; if not, then the customer joins the queue (that is, he waits in line). When the server finishes serving a customer, the customer leaves the system and the next customer in line, if there are any waiting, enters the service. The successive service times are assumed to be independent exponential random variables having mean $1/\mu$.

M / M / 1 → Single server.

- M: interarrival process is Markov because it is Poisson
- M: Service time is exponential. Hence, the process is Markov

$X(t)$ = # of customers in the system at t.

- Time from n to $n+1 \sim Exp(\lambda)$.
- ---- n to $n-1 \sim Exp(\mu)$.



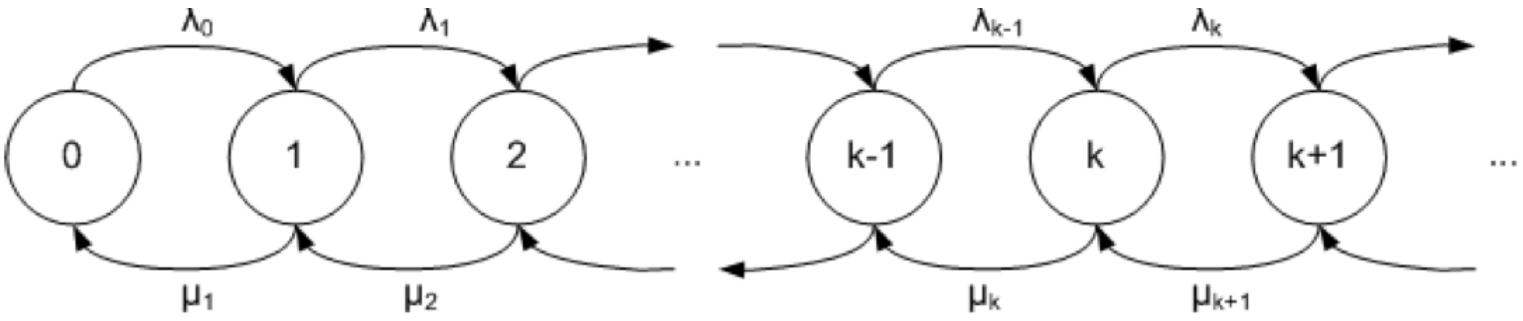
$\lambda_i \equiv \lambda$

$\mu_i \equiv \mu$

Examples

- **Example 6.6 (A Multiserver Exponential Queueing System)** Consider an exponential queueing system in which there are s servers available, each serving at rate μ. An entering customer first waits in line and then goes to the first free server. Assume arrivals are according to a Poisson process having rate λ.

M/M/s



$\lambda_n \equiv \lambda, \ n \geq 0$

$\mu_n \equiv n\mu, \ 1 \leq n \leq s, \quad \mu_n = s\mu, \ n > s.$

Transition probabilities and Q matrix

- In general, to compute the transition probability $P(X(t+s)=j|X(s)=i)$, for general CTMC is difficult. Instead, we shall characterize the transition in infinitesimal time intervals and express the transition probabilities in some differential equations.
- **Lemma 6.2** Define $q_{ij} = v_i P_{ij}$. Then, given $X(0)=i$,

$$\lim_{h\to 0} \frac{1-P_{ii}(h)}{h} = v_i,$$

and for all $j \neq i$,

$$\lim_{h\to 0} \frac{P_{ij}(h)}{h} = q_{ij}.$$

$P_{ij}(h) = P(X(h)=j \mid X(0)=i)$

Proof. Let $\xi \sim \text{Exp}(\nu_i)$ with pdf $\nu_i e^{-\nu_i y}$, $y > 0$.

Then $$\begin{aligned} P_{ii}(h) &= P(X(h) = i \mid X(0) = i) \\ &= P(\xi > h) + o(h) \\ &= e^{-\nu_i h} + o(h). \end{aligned}$$

Thus $$\frac{1 - P_{ii}(h)}{h} \longrightarrow \nu_i$$

For $j \neq i$,

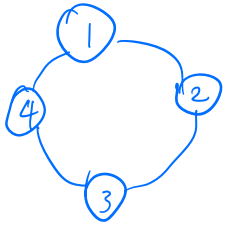
$$\begin{aligned} P_{ij}(h) &= P(X(h) = j \mid X(0) = i) \\ &= P(X(h) = j, \xi < h \mid X(0) = i) \\ &= P_{ij}(1 - e^{-\nu_i h}) \end{aligned}$$

Then $$\frac{P_{ij}(h)}{h} \longrightarrow P_{ij}\nu_i \quad \text{as } h \to 0.$$

Transition probabilities and Q matrix

- We call q_{ij} the rate to transit from state i to state j. The matrix $Q = (q_{ij})$ defined as $q_{ij} = v_i P_{ij}$ if $i \neq j$ and $= -v_i$ if $i = j$ is called the Q-matrix.

- **Example:** Write down the Q-matrix for the rat in the maze.



$P_{ij} = \frac{1}{2}$ for any $i \neq j$.

Holding time at $i \sim \text{Exp}(a_i)$.

$$Q = \begin{pmatrix} -a_1 & \frac{1}{2}a_1 & \frac{1}{2}a_1 & \frac{1}{2}a_1 \\ \frac{1}{2}a_2 & -a_2 & \frac{1}{2}a_2 & \frac{1}{2}a_2 \\ \frac{1}{2}a_3 & \frac{1}{2}a_3 & -a_3 & \frac{1}{2}a_3 \\ \frac{1}{2}a_4 & \frac{1}{2}a_4 & \frac{1}{2}a_4 & -a_4 \end{pmatrix}$$

Lemma 6.3 (Chapman-Kolmogorov eq). For $s, t \geq 0$,

$$P_{ij}(t+s) = \sum_k P_{ik}(t) P_{kj}(s).$$

Proof. $P_{ij}(t+s) = P(X(t+s) = j \mid X(0) = i)$

$$= \sum_k P(X(t+s) = j, \; X(s) = k \mid X(0) = i)$$

But the last prob is

$$P(X(t+s) = j \mid X(t) = k, X(0) = i) \cdot P(X(t) = k \mid X(0) = i)$$

Markov property $\quad P(X(t+s) = j \mid X(t) = k) \cdot P_{ik}(t)$

$$= P_{kj}(s) \cdot P_{ik}(t).$$

Kolmogorov's Backward Equation

- **Theorem 6.1** Let $P_{ij}(t)$ be the transition probability

$$P_{ij}(t) = P(X(t) = j | X(0) = i).$$

Then, for any $i, j \in \mathcal{S}$,

$$P'_{ij}(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - v_i P_{ij}(t). = \sum_{k} q_{ik} P_{kj}(t)$$

Let $P(t) = (P_{ij}(t))$ be the transition matrix, we have

$$P'(t) = QP(t).$$

$P(0) = I$

Proof: $P_{ij}(h+t) - P_{ij}(t)$

$$= \left[\sum_k P_{ik}(h) P_{kj}(t)\right] - P_{ij}(t)$$

$$= \left[\sum_{k \neq i} P_{ik}(h) P_{kj}(t)\right] - P_{ij}(t)(1 - P_{ii}(h)).$$

$$\therefore \frac{P_{ij}(h+t) - P_{ij}(t)}{h}$$

$$= \sum_{k \neq i} \frac{P_{ik}(h)}{h} P_{kj}(t) - \frac{1 - P_{ii}(h)}{h} P_{ij}(t)$$

Take limit $h \to 0$, we get

$$P_{ij}'(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - v_i P_{ij}(t)$$

$$\underline{\underline{Q_{ii} = -v_i}} \sum_k q_{ik} P_{kj}(t).$$

Equivalently, set $P(t) = [P_{ij}(t)]$. Then

$$P'(t) = Q P(t).$$

Example

- **Example 6.11 (A Continuous-Time Markov Chain Consisting of Two States)** Consider a machine that works for an exponential amount of time having mean $1/\lambda$ before breaking down; and suppose that it takes an exponential amount of time having mean $1/\mu$ to repair the machine. If the machine is in working condition at time 0, then what is the probability that it will be working at time $t = 10$?

Solution. Let

0 : the machine is working.

1 : " " " being repaired.

The machine working or not is in the format:

$0 \longrightarrow 1$, $1 \longrightarrow 0$, 2-state MC.

λ_0 = the parameter of the Exp. dist. stays in 0 before moving to 1

$= \lambda$

μ_1 = the parameter of the Exp. dist. stays in 1 before moving to 0

$= \mu$.

$P_{01} = 1$, $P_{10} = 1$,

$P_{00} = 0$, $P_{11} = 0$.

By the Kolmogorov backward eq.

$$P_{00}'(t) = \lambda [P_{10}(t) - P_{00}(t)] \quad (1)$$

$$P_{10}'(t) = \mu [P_{00}(t) - P_{10}(t)] \quad (2)$$

Obviously, $\left[\mu P_{00}(t) + \lambda P_{10}(t)\right]' = 0$

$\therefore \quad \mu P_{00}(t) + \lambda P_{10}(t) \equiv C$

Initial condition: $P_{00}(0) = 1$. $P_{10}(0) = 0$. Thus

$C \equiv \mu$. Thus

$$\mu P_{00}(t) + \lambda P_{10}(t) = \mu. \qquad (3)$$

Equivalently

$$\lambda P_{10}(t) = (1 - P_{00}(t))\mu$$

By substituting this in (1), we get

$$P_{00}'(t) = \mu - (\lambda + \mu) P_{00}(t).$$

By standard trick in ordinary eq:

$$P_{00}(t) = k e^{-(\mu+\lambda)t} + \frac{\mu}{\mu+\lambda}.$$

where k is a const. Use $P_{00}(0) = 1$ again to see

$k = \frac{\lambda}{\mu+\lambda}$. Thus

$$P_{00}(t) = \frac{\lambda}{\mu+\lambda} e^{-(\mu+\lambda)t} + \frac{\mu}{\mu+\lambda}.$$

By substituting in (3), we get

$$P_{10}(t) = \frac{\mu}{\mu+\lambda} - \frac{\mu}{\mu+\lambda} e^{-(\mu+\lambda)t}.$$

Note: $P_{01}(t) = 1 - P_{00}(t)$ & $P_{11}(t) = 1 - P_{10}(t)$.

An implication of Kolmogorov's forward/backward equation

Let $P=(P_{ij})$ be the embedded MC. At each state i, there is an exponential dist. $\text{Exp}(v_i)$. Set $Q=(q_{ij})$, where

$$q_{ij}=\begin{cases} -v_i, & \text{if } i=j; \\ v_i P_{ij}, & \text{if } i\neq j. \end{cases}$$

Recall $P_{ij}(t)=P(X(t)=j \mid X(0)=i)$ is the transition probabilities of the CTMC $X(t)$ based on P and $v_i, i\geq 1$. Write $P(t)=(P_{ij}(t))$. By the Kolmogorov backward/forward eq, $P'(t)=(P_{ij}'(t))=QP(t)=P(t)Q$. Of course, $P(0)=I$. Define $e^A=I+\sum_{n=1}^{\infty}\frac{A^n}{n!}$ for any matrix A. It is easy to check

$$P(t)=e^{Qt} \qquad (*)$$

is the sol of $P'=QP=PQ$. As a result

$$P_{ij}(t)=\text{the } (i,j)\text{-entry of matrix } e^{Qt}.$$

As a comparsion, for discrete MC Y_n with transition matrix P, we have

$$P_{ij}^{(n)}=P(X(n)=j \mid X(0)=i)=(i,j)\text{-entry of matrix } P^n. \qquad (**)$$

Compare $(*)$ & $(**)$, we see the roles of P and e^Q are the same.

- **Example 6.11 (A Continuous-Time Markov Chain Consisting of Two States)** Consider a machine that works for an exponential amount of time having mean $1/\lambda$ before breaking down; and suppose that it takes an exponential amount of time having mean $1/\mu$ to repair the machine. If the machine is in working condition at time 0, then what is the probability that it will be working at time $t = 10$?

Another sol. Recall

0 : the machine is working.

1 : " " " being repaired.

The machine working or not is in the format:

$0 \longrightarrow 1$, $1 \longrightarrow 0$. 2-state MC.

λ_0 = the parameter of the Exp. dist. stays in 0 before moving to 1 $= \lambda$

μ_1 = the parameter of the Exp. dist. stays in 1 before moving to 0 $= \mu$

$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then $Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$.

Check $Q^2 = -(\lambda+\mu)Q$. Hence $Q^n = (-(\lambda+\mu))^{n-1} Q$.

Hence

$$e^{tQ} = I_2 + \sum_{n=1}^{\infty} \frac{Q^n}{n!} t^n$$

$$= I_2 - \frac{Q}{\lambda+\mu} \sum_{n=1}^{\infty} \frac{[-(\lambda+\mu)t]^n}{n!}$$

$$= I_2 - \frac{1}{\lambda+\mu} \left(e^{-(\lambda+\mu)t} - 1\right) Q$$

$P_{00}(t)$ ←

$$= \begin{pmatrix} 1 + \frac{\lambda}{\lambda+\mu}\left(e^{-(\lambda+\mu)t} - 1\right) & -\frac{\lambda}{\lambda+\mu}\left(e^{-(\lambda+\mu)t} - 1\right) \\ -\frac{\mu}{\lambda+\mu}\left(e^{-(\lambda+\mu)t} - 1\right) & 1 + \frac{\mu}{\lambda+\mu}\left(e^{-(\lambda+\mu)t} - 1\right) \end{pmatrix}$$

$P_{10}(t)$ ←

The sol. same as before.

Kolmogorov's Forward Equations

- **Theorem (Informal)** Let $P(t)$ be the transition probability matrix that

$$P_{ij}(t) = P(X(t) = j | X(0) = i).$$

Then, under certain conditions (e.g. finite state space), for all $i, j \in \mathcal{S}$,

$$P'_{ij}(t) = \sum_{k \neq i} q_{kj} P_{ik}(t) - v_j P_{ij}(t).$$

In the matrix form, we have

$$P'(t) = P(t)Q.$$

Proof: $P_{ij}(t+h) - P_{ij}(t)$

$$= \left[\sum_k P_{ik}(t) P_{kj}(h)\right] - P_{ij}(t)$$

$$= \left[\sum_{k \neq j} P_{ik}(t) P_{kj}(h)\right] - P_{ij}(t)(1 - P_{jj}(h))$$

$$\therefore \quad \frac{P_{ij}(h+t) - P_{ij}(t)}{h}$$

$$= \sum_{k \neq j} \frac{P_{kj}(h)}{h} P_{ik}(t) - \frac{1 - P_{jj}(h)}{h} P_{ij}(t)$$

Take limit $h \to 0$, we get

$$P_{ij}'(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - v_j \cdot P_{ij}(t)$$

$$\overset{q_{jj} = -v_j}{=\!=} \sum_k q_{kj} P_{ik}(t).$$

Basically, in the proof of backwasd eq, we swap the roles of "t" and "h" by changing "$P_{ij}(t+h)$" to "$P_{ij}(h+t)$"

Classification of States

- State j is said to be accessible from state i if $P_{ij}(t) > 0$ for some $t > 0$.
- We say state i and j communicate if state j is accessible from i and state i is accessible from j.

- State j is accessible from state i if and only if it is so in the embedded chain.
- Two states i and j communicate in the CTMC if and only if they do so in the embedded DTMC.

- **Proposition:** A CTMC has the same communication classes as its embedded chain. In corollary, a CTMC is irreducible if and only if the embedded chain is irreducible.

Transient / Recurrent States

- Let T_{ii} denote the amount of time until the chain re-visit state i after it departs from i, i.e.

$$T_{ii} = \inf\{t > H_i : X_t = i, \text{ given that} X_0 = i\}.$$

H_i : exponential

- If $P(T_{ii} < \infty) = 1$, we say state i is recurrent, otherwise, we say i is transient.

- **Proposition:** A state i is recurrent/transient if and only if it is so in the embedded chain

Positive Recurrent

- For a recurrent state i, if $E[T_{ii}] < \infty$, we say i is positive recurrent. Otherwise we say it is null recurrent.

- In general, positive recurrence is not equivalent in CTMC and the embedded chain.
- **Proposition:** For a CTMC with finite states, if a state i is positive recurrent in the embedded chain, it is also positive recurrent in the CTMC.

Limiting Probability

- In analogue to DTMC, very often we have the transition probability $P_{ij}(t) \to$ [redacted] for all $i, j \in \mathcal{S}$. as $t \to \infty$

  P_j

- **Theorem:** A finite-state, irreducible CTMC is positive recurrent. Besides, it has a unique limiting probability $P = (P_j)$. P_j is equal to the long-run proportion, i.e.

$$P_j = \frac{E[H_j]}{E[T_{jj}]} > 0.$$

$H_j \sim Exp(v_j)$

Limiting Probability

- **Proposition:** Let Q be the Q-matrix of a CTMC X_t. Suppose X_t has a limiting probability $P = (P_j)$. Then, (P_j) is the solution to the system of equations

$$PQ = 0, \text{ i.e. } P_j v_j = \sum_{i \neq j} P_i q_{ij}, \forall i \in \mathcal{S}$$

and

$$\sum_{j \in \mathcal{S}} P_j = 1.$$

Proof. By the forward equation,

$$P'_{ij}(t) = \sum_{k \neq i} q_{kj} P_{ik}(t) - v_j P_{ij}(t).$$

Use $P_j = \lim_{t \to \infty} P_{ij}(t)$, we know $\lim_{t \to \infty} P'_{ij}(t) = 0$.

Assuming we can interchange "$\lim_{t \to \infty}$" and "$\sum$", we get

$$0 = \sum_{k \neq i} q_{kj} P_k - v_j P_j$$

Also, since $\sum_j P_{ij}(t) = 1$. Take $t \to \infty$ to see

$$\sum_j P_j = 1.$$

Stationary Distribution

- If $X_0 \sim P$, we can check that $X_t \sim P$ for all $t > 0$.
- So, P is also called the stationary distribution of the CTMC.

Proof. Keep in mind that $P_j = \lim_{t\to\infty} P_{ij}(t)$ for any i, j. Then

$$P(X(t)=j) = \sum_k P(X(t)=j \mid X(0)=k)\cdot \underbrace{P(X(0)=k)}_{P_k}$$

$$= \sum_k P_{kj}(t) P_k$$

$$= \sum_k P_{kj}(t) \lim_{s\to\infty} P_{ik}(s)$$

$$= \lim_{s\to\infty} \sum_k P_{kj}(t) P_{ik}(s)$$

$$= \lim_{s\to\infty} P_{ij}(s+t)$$

$$= P_j .$$

For general case, let $0 = t_0 < t_1 < \cdots < t_k$. By Markov's property,

$$P(X(t_1)=j_1,\ X(t_2)=j_2,\ \cdots,\ X(t_k)=j_k) \qquad \text{(*)}$$

$$= P(X(t_k)=j_k \mid X(t_{k-1})=j_{k-1}) \cdots P(X(t_2)=j_1 \mid X(t_1)=j_1)\, P(X(t_1)=j_1)$$

$$= P_{j_{k-1}j_k}(t_k - t_{k-1}) \cdot P_{j_{k-2}j_{k-1}}(t_{k-1}-t_{k-2}) \cdots P_{j_1 j_2}(t_2 - t_1)\, P_{j_1}$$

observe that the above depends on $t_i - t_{i-1}$ only. Then

$$P(X(t_1+s)=j_1,\ X(t_2+s)=j_2,\ \cdots,\ X(t_k+s)=j_k)$$

$$= P(X(t_1)=j_1,\ X(t_2)=j_2,\ \cdots,\ X(t_k)=j_k)$$

for any s with $s > -t_1$.

Examples

- **Example 6.11 (A Continuous-Time Markov Chain Consisting of Two States)** Consider a machine that works for an exponential amount of time having mean $1/\lambda$ before breaking down; and suppose that it takes an exponential amount of time having mean $1/\mu$ to repair the machine. What is the limiting probability that the machine works?

Solution. Let

0 : the machine is working.

1 : " " " being repaired.

The machine working or not is in the format:

$0 \longrightarrow 1$, $1 \longrightarrow 0$. 2-state MC.

λ_0 = the parameter of the Exp. dist. stays in 0 before moving to 1

$= \lambda$

μ_1 = the parameter of the Exp. dist. stays in 1 before moving to 0

$= \mu$.

$P_{01} = 1$, $P_{10} = 1$,

$P_{00} = 0$, $P_{11} = 0$.

We have shown that

$$P_{00}(t) = \frac{\lambda}{\mu+\lambda} e^{-(\mu+\lambda)t} + \frac{\mu}{\mu+\lambda}.$$

and

$$P_{10}(t) = \frac{\mu}{\mu+\lambda} - \frac{\mu}{\mu+\lambda} e^{-(\mu+\lambda)t}.$$

Let $t \to \infty$, then

$$\lim_{t\to\infty} P_{i0}(t) = \frac{\mu}{\mu+\lambda} = P_0, \quad i = 0, 1.$$

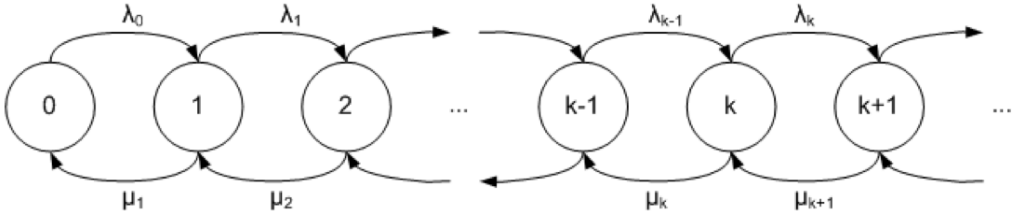
and $P_1 = 1 - P_0 = \frac{\lambda}{\mu+\lambda}$

Limiting probabilites for birth-death processes

For a cont. time discrete state MC $X(t)$, its transition probab. $P_{ij}(t) = P(X(t)=j \mid X(0)=i)$. Its embedded chain has transition matix $P=(P_{ij})$. A sufficient condition for $\lim_{t\to\infty} P_{ij}(t) = P_j$ is that

- the embedded chain is irreducible;
- $X(t)$ is positive recurrent, i.e., starting from any state i, the mean time to return to that state is finite.

Now consider a birth-death process:



We are interested in the restrictions on of λ_i and μ_i given the chain has limiting probabilities $P_0, P_1, \ldots$

By an earlier proposition, such P_i has to satisfy

$$P_0 + P_1 + \cdots = 1$$

$$P_j \nu_j = \sum_{i \neq j} P_i q_{ij}, \quad \forall j.$$

We also showed

$$P_{i,i-1} = \frac{\mu_i}{\mu_i + \lambda_i}, \quad P_{i,i+1} = \frac{\lambda_i}{\mu_i + \lambda_i}, \quad \nu_i = \lambda_i + \mu_i, \; i \geq 1, \; \nu_0 = \lambda_0.$$

Thus $q_{ij} = \nu_i P_{ij} = \begin{cases} \mu_i, & \text{if } j = i-1; \\ \lambda_i, & \text{if } j = i+1. \end{cases}$

We need

$$\lambda_0 P_0 = \mu_1 P_1$$

$$(\lambda_1 + \mu_1) P_1 = \lambda_0 P_0 + \mu_2 P_2$$

$$(\lambda_2 + \mu_2) P_2 = \lambda_1 P_1 + \mu_3 P_3$$

$$\vdots$$

$$(\lambda_n + \mu_n) P_n = \lambda_{n-1} P_{n-1} + \mu_{n+1} P_{n+1}$$

From the 1st 2 eq's, we see

$$\lambda_0 P_0 = \mu_1 P_1$$

$$\lambda_1 P_1 = \mu_2 P_2$$

$$\lambda_n P_n = \mu_{n+1} P_{n+1}, \quad n \geq 0.$$

Solve:

$$P_1 = \left(\frac{\lambda_0}{\mu_1}\right) P_0$$

$$P_2 = \left(\frac{\lambda_1}{\mu_2}\right) P_1$$

$$\vdots$$

$$P_n = \left(\frac{\lambda_{n-1}}{\mu_n}\right) P_1$$

Multiply them:

$$P_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} P_0, \quad n \geq 1.$$

In order to have $P_0 + P_1 + P_2 + \cdots = 1$, we have

$$P_0 = \left(1 + \sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n}\right)^{-1}.$$

As a consequence

$$P_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \left(1 + \sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n}\right)^{-1}, \quad n \geq 1.$$

Thus, the restriction on λ_i, μ_i is

$$\sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} < \infty.$$

Examples

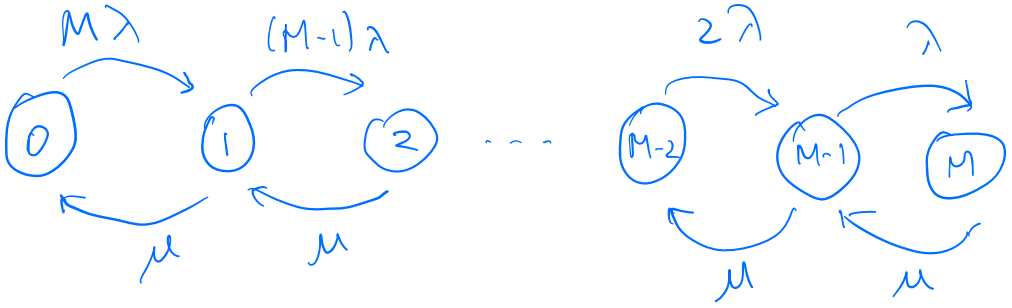
- **Example 6.13 (A Machine Repair Model)** Consider a job shop that consists of M machines and one serviceman. Suppose that the amount of time each machine runs before breaking down is exponentially distributed with mean $1/\lambda$, and suppose that the amount of time that it takes for the serviceman to fix a machine is exponentially distributed with mean $1/\mu$. We shall attempt to answer these questions:

  (a) What is the average number of machines not in use?

  (b) What proportion of time is each machine in use?

  (c) What is the proportion of time a given machine is working?

Solution. This is a birth-death chain:



$\lambda_n = (M-n)\lambda, \quad 0 \le n \le M$

$\mu_n \equiv \mu.$

$X(t) =$ number of machine not in use up to time t.

Now we determine the limiting probabilities:

$$\frac{1}{P_0} = 1 + \sum_{n=1}^{\infty} \frac{\lambda_0 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n}$$

$$= 1 + \sum_{n=1}^{M} \frac{M(M-1)\cdots(M-n+1)\lambda^n}{\mu^n}$$

$$\therefore P_0 = \left(1 + \sum_{n=1}^{M} \frac{M!}{(M-n)!}\left(\frac{\lambda}{\mu}\right)^n\right)^{-1}.$$

$$P_n = \frac{M!}{(M-n)!}\left(\frac{\lambda}{\mu}\right)^n P_0, \quad n \ge 0.$$

(a). Average number of machine not in use

$$= \sum_{n=1}^{M} n P_n .$$

(b) Proportion of time each machine in use

$$= P_0 .$$

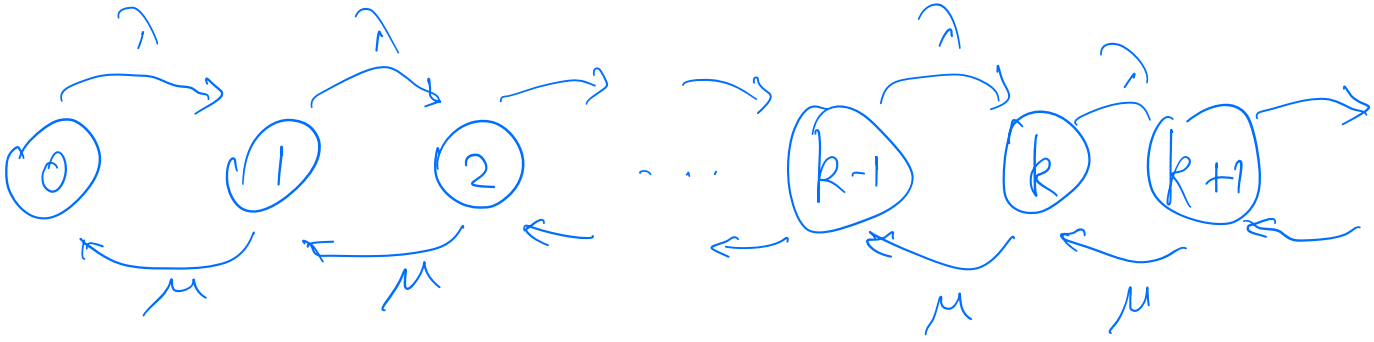
(C) P(Machine 1 is working)

$$= \sum_{n=0}^{M} P(\text{Machine 1 is working} \mid \underset{\text{not working}}{n \text{ machine}}) \cdot P_n$$

$$= \sum_{n=0}^{M} \frac{M-n}{M} P_n .$$

Basic Queueing Models

- M/M/1 queue
- Arrival rate λ, service rate μ.
- In order to have stationary distribution, what condition should λ and μ satisfy?



$$P_0 = \left(1 + \sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n}\right)^{-1}.$$

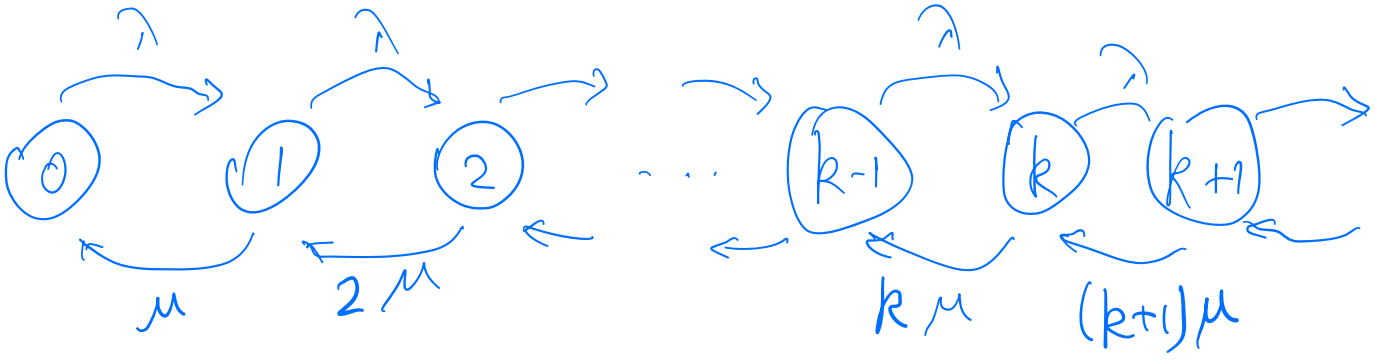
$$= \left(1 + \sum_{n=1}^{\infty} \left(\frac{\lambda}{\mu}\right)^n\right)^{-1}.$$

Thus $\lambda < \mu$. In this case,

$$P_n = \left(\frac{\lambda}{\mu}\right)^n \frac{1}{1 - \frac{\lambda}{\mu}} = \frac{\mu}{\mu - \lambda} \left(\frac{\lambda}{\mu}\right)^n.$$

Basic Queueing Models

- M/M/∞ queue
- Arrival rate λ, service rate μ.
- In order to have stationary distribution, what condition should λ and μ satisfy?



$\lambda_n \equiv \lambda, \quad \mu_n = n\mu, \quad n \geq 0.$

$$P_0 = \left(1 + \sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n}\right)^{-1}.$$

$$= \left(1 + \sum_{n=1}^{\infty} \frac{(\lambda/\mu)^n}{n!}\right)^{-1}$$

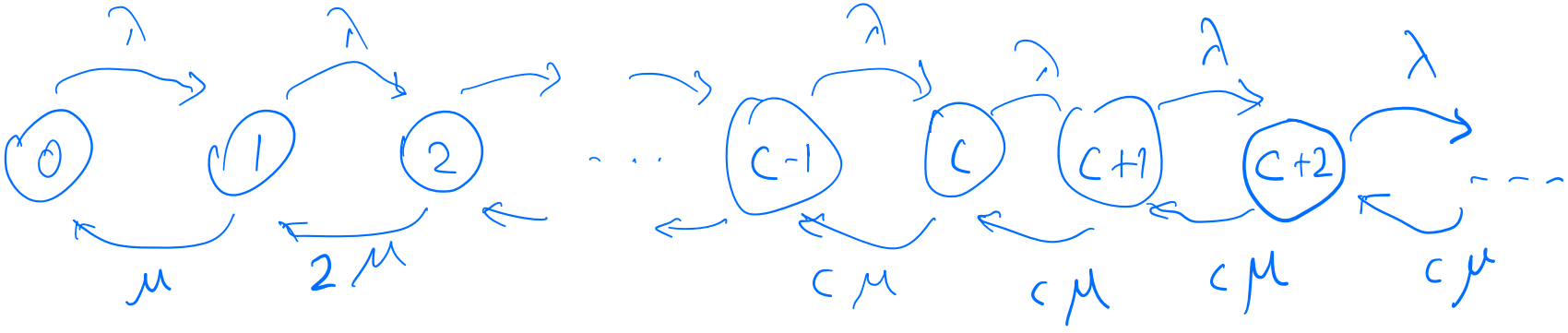
$$= e^{-\lambda/\mu}$$

$$P_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n} P_0 = \frac{(\lambda/\mu)^n}{n!} e^{-\lambda/\mu}.$$

Thus, $P_n \sim \text{Poi}(\alpha)$, $\alpha = \frac{\lambda}{\mu}$.

Basic Queueing Models

- M/M/c queue
- Arrival rate λ, service rate μ.
- In order to have stationary distribution, what condition should λ and μ satisfy?



$$\lambda_i \equiv \lambda, \quad i \geq 0,$$

$$\mu_i = \begin{cases} i\mu, & i \leq c \\ c\mu, & i > c. \end{cases}$$

$$\frac{1}{P_0} = 1 + \sum_{n=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n}$$

As $n \geq c$, we have

$$\frac{\lambda_0 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n} = \frac{1}{c!} \left(\frac{\lambda}{\mu}\right)^c \cdot \frac{\lambda^{n-c}}{(c\mu)^{n-c}}$$

$$= \frac{1}{c!} (\lambda c)^c \cdot \left(\frac{\lambda}{c\mu}\right)^n.$$

We need $\lambda < c\mu$.

Embedded Chain

- Suppose $X(t)$ is a CTMC with Q-matrix Q, let $Y(n)$ be its embedded chain. We know that $Y(n)$ has transition probabilities

$$P_{ij} = q_{ij}/v_i.$$

- Suppose π and π^d are the limiting probabilities of the $X(t)$ and $Y(n)$ respectively. Then,

$$\pi_i = \frac{\pi_i^d/v_i}{\sum_j \pi_j^d/v_j}.$$

  Equivalently,

$$\pi_i^d = \frac{\pi_i v_i}{\sum_j \pi_j v_j}.$$

- If we look the CTMC backwards in time, the amount of time t that the chain stay in state i is still an exponential random variable with mean $1/v_i$.

Time Reversibility

- A CTMC is said to be time reversible if its embedded DTMC is time reversible.
- A CTMC is time reversible if and only if

$$P_i q_{ij} = P_j q_{ji}.$$

$$q_{ij} = \begin{cases} -v_i, & \text{if } i=j \\ v_i p_{ij}, & \text{if } i \neq j \end{cases}$$

- **Proposition 6.5** A stationary birth and death process is time reversible.

$P=(p_{ij})$ is transition matrix of embedded chain

Proof of Prop. 6.5. Recall

$$P_{i\,i-1} = \frac{\mu_i}{\mu_i+\lambda_i}, \quad P_{i\,i+1} = \frac{\lambda_i}{\mu_i+\lambda_i}, \quad \nu_i = \lambda_i+\mu_i, \ i\geq 1, \ \nu_0 = \lambda_0.$$

$$\lambda_n P_n = \mu_{n+1} P_{n+1}, \ n\geq 0 \quad \text{and} \quad q_{ij} = \nu_i P_{ij}.$$

Therefore, the embedded chain is reversible, by definition, if

$$\begin{cases} \pi_i P_{i\,i+1} = \pi_{i+1} P_{i+1\,i} \\ \sum_{i=1}^{\infty} \pi_i = 1 \end{cases}$$

The above is equivalent to

$$\pi_i \frac{\lambda_i}{\mu_i+\lambda_i} = \pi_{i+1} \frac{\mu_{i+1}}{\mu_{i+1}+\lambda_{i+1}}$$

Set

$$P_i = \frac{\pi_i}{\mu_i+\lambda_i}\left(\sum_{i=0}^{\infty} \frac{\pi_i}{\mu_i+\lambda_i}\right)^{-1}.$$

The above implies $\lambda_i P_i = \mu_{i+1} P_{i+1}$.

Since $q_{i\,i+1} = \nu_i P_{i\,i+1} = \lambda_i$, $q_{i+1\,i} = \nu_{i+1} P_{i+1\,i} = \mu_i$, then

$$P_i q_{ij} = P_j q_{ji} \quad \text{for } i\neq j.$$

Examples

- **Example 6.17** Consider a first come first serve M/M/1 queue, with arrival rate λ and service rate μ, where $\lambda < \mu$, that is in steady state. Given that customer C spends a total of t time units in the system, what is the conditional distribution of the number of others that were present when C arrived?

Examples

- **Example (Two tandem queue)** Customers arrive at a service station, manned by a single server who serves at an exponential rate μ_1, at a Poisson rate λ. After completion of service the customer then joins a second system where the server serves at an exponential rate μ_2. Assuming $\lambda < \mu_i$, $i = 1, 2$, determine the limiting probabilities.

Chapter Summary

- Definition of CTMC
  - Exponential holding times
  - Embedded chain (DTMC)
- Birth-death process
- Transition probabilities
  - Q-matrix; transition rates
  - Kolmogorov forward/backward equations
- State classification
  - accessible, communicate
  - transient, recurrent, positive/null recurrent
  - irreducible
- Limiting probability / stationary distribution
  - solve P via linear equations
- Time reversibility